Cartoon Caption Matching and Humor Ranking

CIS 4300/5300 - Milestone 2
Alex Yang, Maggie Du, Celia Tung, Chinmay Govind

1. Tasks and data

We evaluate cartoon humor understanding through two classification tasks: selecting the caption written for a cartoon from five candidates, and choosing which of two captions received a higher crowd rating. We preserve the team's Milestone 1 JSONL splits. Matching uses Hessel et al. [1], split 0. Ranking uses Zhang et al. [2] crowd ratings and the team's fixed pairs (seed 5300).

Task	Train items (contests)	Dev items (contests)	Test items (contests)
Matching	4,896 (422)	531 (141)	528 (141)
Ranking	5,420 (271)	880 (44)	940 (47)

Table 1. Every task is contest-disjoint across splits. Milestone 1 supplies all dev/test images; omitted training images are unnecessary for frozen zero-shot inference.

Easy ranking pairs compare ranks 1-10 with 1000-1009; hard pairs compare ranks 30-39 with 300-309. Each contest supplies ten pairs of each tier, with randomized A/B order. Labels express the crowd's preferences. They need not represent universal taste.

2. Evaluation measure

For N gold examples with labels y_i and system predictions $h(x_i)$, we use exact-match accuracy [1]:

$\text{Acc}(D) = (1/N) \sum_{i=1}^{N} 1[h(x_i) = y_i]$.

We report matching accuracy, ranking accuracy, and accuracy within each ranking tier. The scorer aligns examples by unique ID and rejects missing, extra, or duplicate IDs. Unparsable model responses become null predictions and count as wrong; their count is reported separately. Uniform-random expectations are 20% for matching and 50% for ranking, rather than empirical measurements.

3. Simple baseline

The baseline predicts the most frequent training answer position for every example: **$c^* = \text{argmax}_c \sum_{i \text{ in train}} 1[y_i = c]$; $h(x) = c^*$.** Alphabetical order resolves ties. It predicts E for matching (1,031/4,896 train labels) and B for ranking (2,760/5,420). It reads no evaluation labels to choose its class.

Simple baseline	Dev correct / N	Dev %	Test correct / N	Test %
Matching	105 / 531	19.77	97 / 528	18.37
Ranking overall	431 / 880	48.98	470 / 940	50.00
Ranking easy	202 / 440	45.91	239 / 470	50.85
Ranking hard	229 / 440	52.05	231 / 470	49.15

Table 2. Actual majority-baseline scores from score.py. All predictions are valid. Tier differences reflect label-position frequencies, not humor comprehension.

4. Strong baseline: zero-shot Qwen3.5-4B

We implement a zero-shot multiple-choice vision-language baseline using the official post-trained Qwen3.5-4B checkpoint [3]. It receives the raw cartoon and candidate captions and is asked to return one answer letter. This adapts the caption-contest evaluation approach in [1] to a Chinese-developed open-weight model from our proposal. It does not reproduce the fine-tuned CLIP system in [1].

The model is frozen, with no demonstrations or task training. Input prompts include only the image, task instruction, and captions. Gold labels, crowd means, ranks, tiers, IDs, and scene annotations are excluded. We use greedy decoding, thinking disabled, at most 16 generated tokens, one example per batch, and a maximum image area of 512 x 512 pixels. Only an exact uppercase answer letter is accepted. Model revisions, input hashes, GPU type, precision, and package versions are recorded.

Qwen3.5-4B (pixels)	Dev accuracy %	Test accuracy %
Matching	47.27 (251/531)	49.62 (262/528)
Ranking overall	50.80 (447/880)	51.49 (484/940)
Ranking easy	48.41 (213/440)	52.77 (248/470)
Ranking hard	53.18 (234/440)	50.21 (236/470)

Table 3. Strong-baseline results are computed from complete prediction files. Counts in parentheses give correct predictions / total examples.

The completed runs used NVIDIA L4 with torch.bfloat16. Model commit: 851bf6e806efd8d0a36b00ddf55e13ccb7b8cd0a.

The four strong-baseline runs produced 0 unparsable responses in total; all are included as errors. The prediction files retain raw responses for review.

Relative to the simple baseline, test accuracy changed by +31.25 percentage points for matching and +1.49 points for ranking.

Ranking remained close to chance. Qwen chose B on 826/940 test pairs (87.87%), although gold A/B labels were exactly balanced. This limits the evidence for crowd-preference understanding.

A separate dev-only order-swap diagnostic preserved the cartoon and captions but exchanged A/B positions and flipped gold labels. Qwen kept the same caption on only 159/880 pairs (18.07%); it chose B in both orders on 721/880 pairs (81.93%), changing the chosen caption. Dev accuracy changed from 50.80% to 53.41%. This supports strong order sensitivity in this setup; it is not a calibrated model or a new test result. A related GPT-4 position bias is discussed in [2, Appendix C.2].

5. Interpretation and limitations

The matching paper reports five-fold averages; our single-fold scores are not directly comparable. Our deterministic Zhang ranking pairs also differ from other papers' possible samples and conditions. Multiple items share cartoons, so uncertainty estimates should resample contests rather than independent items. Pretraining exposure to public cartoons is unknown. A model's origin alone cannot establish a cultural explanation for differences in crowd agreement.

References and implementation attribution

[1] Hessel et al. (2023). Do Androids Laugh at Electric Sheep? Humor "Understanding" Benchmarks from The New Yorker Caption Contest. ACL. aclanthology.org/2023.acl-long.41

[2] Zhang et al. (2024). Humor in AI: Massive Scale Crowd-Sourced Preferences and Benchmarks for Cartoon Captioning. NeurIPS Datasets and Benchmarks. arxiv.org/abs/2406.10522

[3] Qwen Team (2026). Qwen3.5: Towards Native Multimodal Agents. huggingface.co/Qwen/Qwen3.5-4B. Original integration scripts use Transformers and the team's Milestone 1 data. AI assistance was used in drafting code and documentation.